



UNITED INTERNATIONAL UNIVERSITY
Department of Computer Science and Engineering (CSE)
Course Syllabus

1	Course Title	Database Management Systems
2	Course Code	CSE 3521/CSI 221
3	Trimester and Year	Summer 2025
4	Pre-requisites	NIL
5	Credit Hours	3.00
6	Section	D, P
7	Class Hours	D (Sat, Tue – 12:31 PM – 1:50 PM) P (Sun, Wed – 1:51 PM – 3:10 PM)
8	Classroom	D (402) P (403)
9	Course Teacher	Md. Enamul Haque
10	Email	enamul.m3h@gmail.com
11	Office	935(B)
12	Counselling Hours	Check ELMS
13	Text Book	1. Database System Concepts (7th Edition) by Abraham Silberschatz, Henry F. Korth and S. Sudarshan 2. Database Systems: The Complete Book by Garcia-Molina, Ullman and Widom
14	Reference	http://www.db-book.com/
15	Course Contents (approved by UGC)	Concepts and methods in database system, File organization and retrieval, Data manipulation, Query formulation and language, Database models, Data description languages, database integrity and security, Data dictionary/directory systems, database administration, Database design, Survey of some existing database management systems, Some applications using commercial languages.
17	Teaching Methods	Lecture (L), Case Study (CS), Q/A, Assignment (A), Class Test (CT), Mid, Final exam

18	Course Assessments			
		CO	Assessment Method	(%)
		-	Attendance	5
		-	Assignments	5
		-	Class Tests	20
		CO1, CO2	Midterm exam	30
		CO3, CO4	Final exam	40
20	Lecture Outline			

Class	Topics/Assignments	COs	Reading Reference	Lecture Outcomes/Activities
1	Introduction to database, relational database, necessity of database management systems. Different data models and administration of database.	CO1, CO2	Sec 1.2, 1.3	Student will know the importance of database management systems
2	Relational Algebra: Basic Operations	CO2	Sec 2.6	Understand the internal process of database queries
3	Relational Algebra: complex Operations	CO2	Sec 2.6	Understand the internal process of database queries
4	Relational Database: security and integrity management	CO2	Sec 4.3, 4.4	Impose the security and integrity in database.
5	Introduction to schema models, definition and manipulation language,	CO1	Sec 1.4	Introduce with the schema concepts of database
6	Relational Database: constraints, attributes, manipulations.	CO1, CO2	Sec 3.2, 3.9	Introduce with relational database model
7	Relational Database: simple query details	CO1, CO2	Sec 3.2, 3.3, 3.4	Apply basic queries using enterprise language
8	Relational Database: complex query details	CO1, CO2	Sec 3.7, 3.8, 4.1	Apply complex queries using enterprise language
9	Relational Database: View concepts	CO1, CO2	Sec 4.2	Visualize the outcome of database queries
10	Entity relationship data model: attributes and constraints.	CO1	Sec 6.2, 6.3, 6.4	Get the idea of entity database model
11	Entity relationship data model: case study to design database.	CO1, CO2	Sec 6.5, 6.7	Apply the knowledge of entity model to design real life database
12	Review of mid syllabus	--	--	

		MIDTERM EXAM			
13	Database theories: functional dependencies, impact of functional dependencies.	C03	Sec 7.4	Know the theories of functional dependencies	
14	Database techniques: anomaly problems, normalization, different normal forms	C03	Sec 7.3	Know the techniques to minimize redundancy	
15	Database theories and techniques: application of functional dependencies to normalize the database into different normal forms.	C03	Sec 7.5	Apply the knowledge of database theories to reduce redundancy	
16	Application of Normalization in designing database on real life problems	C03, C04	Sec 7.5	Apply the database theory and techniques in real life.	
17	Indexing and Hashing: introduction, importance, types, applications	C03	Sec 14.1, 14.2	Understand the database techniques to reduce the time complexity of queries	
18	Indexing: B+ tree structure, manipulation of B+ tree structure	C03	Sec 14.3	Understand the database techniques to reduce the time complexity of queries	
19	Hashing: Dynamic hash structure and its manipulation	C03	Sec 14.5	Understand the database techniques to reduce the time complexity of queries	
20	Transaction: definition, characteristics, importance, states	C03	Sec 17.1, 17.4	Map the database knowledge with real tasks	
21	Transaction: consistency and serializability	C03	Sec 17.6	Impose the important characteristics to ensure actual tasks	
22	Transaction: atomicity and back up system. RAID: different levels.	C04	Sec 12.5	Understand the back up techniques.	
23	File storage management	C04	Sec 12.1	Visualize the overall file systems.	
24	Review of final syllabus	--	--		

Appendix 2: Grading Policy

Letter Grade	Marks %	Grade Point	Letter Grade	Marks%	Grade Point
A (Plain)	90-100	4.00	C+ (Plus)	70-73	2.33

A- (Minus)	86-89	3.67	C (Plain)	66-69	2.00
B+ (Plus)	82-85	3.33	C- (Minus)	62-65	1.67
B (Plain)	78-81	3.00	D+ (Plus)	58-61	1.33
B- (Minus)	74-77	2.67	D (Plain)	55-57	1.00
			F (Fail)	<55	0.00

Appendix-3: Program outcomes

PO1	An ability to apply knowledge of mathematics, science, and engineering
PO2	An ability to identify, formulate, and solve complex engineering problems
PO3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
PO4	An ability to investigate complex problems using research-based knowledge and research methods design and conduct experiments, as well as to analyze and interpret data
PO5	An ability to use the techniques, skills, and modern engineering tools necessary for engineering practice
PO6	The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental, and societal context
PO7	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts
PO8	An understanding of professional and ethical responsibility
PO9	An ability function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings
PO10	An ability to communicate effectively
PO11	Project management and finance
PO12	A recognition of the need for, and an ability to engage in life-long learning

